

INTRODUCTION

60°, 300° L.H.A.

LATITUDE SAME NAME AS DECLINATION

N. Lat. { L.H.A. greater than 180°Zn=Z
L.H.A. less than 180°Zn=360°-Z

	15°			16°			17°			18°			19°			20°			21°			22°			
Dec.	Hc	d	Z	Hc	d	Z	Hc	d	Z	Hc	d	Z	Hc	d	Z	Hc	d	Z	Hc	d	Z	Hc	d	Z	Dec.
0	28 52.7	+17.5	98.5	28 43.6	+18.6	99.0	28 33.9	+19.7	99.6	28 23.6	+20.9	100.1	28 12.8	+21.9	100.6	28 01.5	+23.0	101.2	27 49.6	+24.0	101.7	27 37.1	+25.2	102.2	0
1	29 10.2	+16.9	97.4	29 02.2	+18.1	98.0	28 53.6	+19.2	98.5	28 44.5	+20.3	99.0	28 34.7	+21.5	99.6	28 24.5	+22.5	100.1	28 13.6	+23.7	100.7	28 02.3	+24.7	101.2	1
2	29 27.1	+16.4	96.3	29 20.3	+17.5	96.9	29 12.8	+18.7	97.4	29 04.8	+19.8	98.0	28 56.2	+20.9	98.5	28 47.0	+22.0	99.1	28 37.3	+23.1	99.6	28 27.0	+24.2	100.1	2
3	29 43.5	+15.9	95.2	29 37.8	+17.0	95.8	29 31.5	+18.1	96.3	29 24.6	+19.2	96.9	29 17.1	+20.4	97.4	29 09.0	+21.5	98.0	29 00.4	+22.6	98.6	28 51.2	+23.7	99.1	3
4	29 59.4	+15.2	94.1	29 54.8	+16.4	94.7	29 49.6	+17.6	95.2	29 43.8	+18.8	95.8	29 37.5	+19.9	96.4	29 30.5	+21.0	96.9	29 23.0	+22.1	97.5	29 14.9	+23.2	98.0	4
5	30 14.6	+14.7	93.0	30 11.2	+15.9	93.5	30 07.2	+17.0	94.1	30 02.6	+18.2	94.7	29 57.4	+19.3	95.3	29 51.5	+20.5	95.9	29 45.1	+21.6	96.4	29 38.1	+22.7	97.0	5
6	30 29.3	+14.1	91.8	30 27.1	+15.3	92.4	30 24.2	+16.5	93.0	30 20.8	+17.6	93.6	30 16.7	+18.8	94.2	30 12.0	+19.9	94.8	30 06.7	+21.1	95.3	30 00.8	+22.2	95.9	6
7	30 43.4	+13.6	90.7	30 42.4	+14														3	+20.5	94.3	30 23.0	+21.7	94.8	7
8	30 57.0	+12.9	89.6	30 57.1	+14														3	+20.0	93.2	30 44.7	+21.1	93.8	8
9	31 09.9	+12.3	88.4	31 11.2	+13														3	+19.4	92.1	31 05.8	+20.6	92.7	9
10	31 22.2	+11.7	87.3	31 24.7	+12														7	+18.8	91.0	31 26.4	+20.0	91.6	10
11	31 33.9	+11.1	86.1	31 37.6	+12														5	+18.3	89.8	31 46.4	+19.4	90.5	11
12	31 45.0	+10.5	85.0	31 49.9	+11														3	+17.6	88.7	32 05.8	+18.8	89.4	12
13	31 55.5	+9.8	83.8	32 01.6	+11														1	+17.1	87.6	32 24.6	+18.2	88.2	13
14	32 05.3	+9.2	82.7	32 12.7	+10														5	+16.4	86.5	32 42.8	+17.7	87.1	14
15	32 14.5	+8.6	81.5	32 23.1	+9														3	+15.8	85.3	33 00.5	+16.9	86.0	15
16	32 23.1	+7.8	80.3	32 32.8	+9														7	+15.1	84.2	33 17.4	+16.4	84.8	16
17	32 30.9	+7.3	79.2	32 41.9	+8														3	+14.5	83.0	33 33.8	+15.7	83.7	17
18	32 38.2	+6.6	78.0	32 50.4	+7														3	+13.9	81.8	33 49.5	+15.1	82.5	18
19	32 44.8	+5.9	76.8	32 58.1	+7														2	+13.2	80.7	34 04.6	+14.4	81.3	19
20	32 50.7	+5.2	75.6	33 05.2	+6														1	+12.5	79.5	34 19.0	+13.7	80.2	20
21	32 55.9	+4.6	74.4	33 11.7	+5														3	+11.8	78.3	34 32.7	+13.0	79.0	21
22	33 00.5	+3.8	73.2	33 17.4	+5														7	+11.1	77.1	34 45.7	+12.3	77.8	22
23	33 04.3	+3.3	72.0	33 22.5	+4														3	+10.4	75.9	34 58.0	+11.7	76.6	23
24	33 07.6	+2.5	70.9	33 26.9	+3														2	+9.7	74.7	35 09.7	+10.9	75.4	24
25	33 10.1	+1.8	69.7	33 30.6	+3														3	+9.0	73.5	35 20.6	+10.2	74.2	25
26	33 11.9	+1.2	68.5	33 33.6	+2														3	+8.3	72.3	35 30.8	+9.5	73.0	26
27	33 13.1	+0.4	67.3	33 36.0	+1														2	+7.5	71.1	35 40.3	+8.7	71.8	27
28	33 13.5	-0.2	66.1	33 37.6	+0														7	+6.8	69.9	35 49.0	+8.0	70.6	28
29	33 13.3	-0.9	64.9	33 38.5	+0														5	+6.1	68.7	35 57.0	+7.3	69.3	29
30	33 12.4	-1.6	63.7	33 38.7	-0														5	+5.3	67.4	36 04.3	+6.5	68.1	30
31	33 10.8	-2.2	62.5	33 38.3	-1														3	+4.5	66.2	36 10.8	+5.7	66.9	31
32	33 08.6	-3.0	61.3	33 37.1	-1														1	+3.8	65.0	36 16.5	+5.0	65.6	32
33	33 05.6	-3.6	60.1	33 35.3	-2														2	+3.1	63.7	36 21.5	+4.2	64.4	33
34	33 02.0	-4.3	58.9	33 32.7	-3														3	+2.3	62.5	36 25.7	+3.4	63.2	34
35	32 57.7	-5.0	57.7	33 29.5	-4														5	+1.5	61.3	36 29.1	+2.7	61.9	35
36	32 52.7	-5.7	56.5	33 25.5	-4														1	+0.8	60.0	36 31.8	+1.9	60.7	36
37	32 47.0	-6.3	55.4	33 20.9	-5														3	0.0	58.8	36 33.7	+1.1	59.4	37
38	32 40.7	-7.0	54.2	33 15.6	-6														3	-0.7	57.6	36 34.8	+0.3	58.2	38
39	32 33.7	-7.6	53.0	33 09.6	-6														2	-1.6	56.3	36 35.1	-0.4	56.9	39
40	32 26.1	-8.3	51.8	33 03.0	-7														5	-2.2	55.1	36 34.7	-1.2	55.7	40
41	32 17.8	-9.0	50.6	32 55.6	-8														1	-3.1	53.9	36 33.5	-2.0	54.5	41
42	32 08.8	-9.6	49.5	32 47.6	-8														3	-3.7	52.6	36 31.5	-2.7	53.2	42
43	31 59.2	-10.2	48.3	32 39.0	-9														3	-4.6	51.4	36 28.8	-3.6	52.0	43
44	31 49.0	-10.8	47.1	32 29.6	-9														3	-5.3	50.2	36 25.2	-4.3	50.7	44
45	31 38.2	-11.5	46.0	32 19.7	-10														7	-6.0	48.9	36 20.9	-5.1	49.5	45
46	31 26.7	-12.1	44.8	32 09.1	-11														7	-6.8	47.7	36 15.8	-5.8	48.3	46
47	31 14.6	-12.7	43.7	31 57.8	-11														7	-7.5	46.5	36 10.0	-6.6	47.0	47
48	31 01.9	-13.3	42.6	31 45.9	-12														1	-8.2	45.3	36 03.4	-7.3	45.8	48
49	30 48.6	-13.9	41.4	31 33.4	-13														2	-9.0	44.1	35 56.1	-8.1	44.6	49
50	30 34.7	-14.5	40.3	31 20.3	-13														3	-9.7	42.9	35 48.0	-8.8	43.3	50
51	30 20.2	-15.1	39.2	31 06.6	-14.3																				